

Joshua Nyandwaki

 **Home :** Kenya

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Gender: Male

ABOUT ME

A passionate Mechatronic Engineer specialized in Automation and Robotics, with hands-on experience in robotics integration and industrial automation implementation. Skilled in designing, implementing, and optimizing automated control and electrical systems bridging the gap between mechanical, electrical and digital technologies.

WORK EXPERIENCE

Synkron International

[07/04/2025 – 03/2026]

Mechatronic and Industrial Automation Engineering Trainee

Supported the design, installation, programming, commissioning, and maintenance of mechatronic, electrical, and industrial automation systems, ensuring the reliable and efficient operation of production equipment. Worked with PLCs, HMIs, SCADA systems, VFDs, electric motors, electrical control panels, pneumatic systems, and compressed air equipment to support manufacturing and automation processes.

- Programmed and configured PLCs, HMIs, SCADA systems, sensors, and industrial control devices.
- Installed, configured, and troubleshoot VFDs and motor control systems.
- Assembled, wired, tested, and maintained electrical control panels using components such as MCCBs, MCBs, contactors, and overload relays.
- Assisted in the installation, commissioning, and testing of automated machinery and production lines.
- Performed preventive and corrective maintenance on electro-mechanical, electrical, pneumatic, and automation equipment to minimise downtime and improve reliability.
- Supported troubleshooting and maintenance of pneumatic systems and air compressors.
- Prepared technical reports, maintenance records, and technical documentation.
- Collaborated with cross-functional teams to optimise processes and improve system efficiency.

CENTRE FOR DEVELOPMENT OF ELECTRONIC DEVICES

[07/08/2022 – 21/06/2024]

Mechatronic engineering Technician

Supported the design, prototyping, and testing of mechatronic and embedded systems, contributing to research and development projects involving robotics, IoT, and electronic product development.

- Participated in the design, assembly, programming, and testing of a robotic arm prototype using the Robot Operating System (ROS), integrating mechanical, electrical, and control systems.
- Developed CAD models and utilised 3D printing technologies to fabricate and refine components for robotic and prototype applications.
- Developed and tested PCBs for smart water metering systems, supporting circuit design, component integration, and system validation.
- Designed and implemented IoT-based embedded systems, including a fingerprint attendance and clocking device.
- Programmed and configured microcontrollers and integrated sensors, actuators, and communication modules for embedded and prototype development.
- Conducted troubleshooting, diagnostics, and functional testing on embedded and prototype systems to identify and resolve technical issues.

- Supported rapid prototyping activities and documented design processes, test results, and technical reports.
- Gained hands-on experience in robotics, ROS, embedded systems development, electronics prototyping, CAD modelling, additive manufacturing, IoT, and product development.

Advanced Design Manufacturing and Training Centre (ADMATC)

[06/02/2022 – 29/07/2022]

Mechanical Engineering Intern – Machining & Fabrication

Supported manufacturing and fabrication operations through CNC machining, CAM programming, welding, and precision inspection, ensuring components were produced to specification and quality standards.

- Set up, programmed, operated, and maintained CNC machines for the production of precision components.
- Utilised Siemens NX for CAM programming and machining preparation to support CNC manufacturing processes.
- Performed manual machining operations, including turning, milling, drilling, and tapping.
- Conducted arc welding, fabrication, and assembly of cabinets and mechanical structures in accordance with engineering drawings and specifications.
- Interpreted technical drawings, blueprints, and manufacturing specifications to support machining and fabrication activities.
- Inspected components using precision measuring instruments such as callipers, micrometres, and dial indicators to verify dimensional accuracy and quality.
- Assisted in troubleshooting and optimising machining and fabrication processes to improve efficiency and product quality.
- Maintained machine tools and workshop equipment through routine cleaning, lubrication, calibration, and preventive maintenance.
- Ensured compliance with safety procedures, workshop best practices, and quality standards.
- Prepared production records, inspection reports, and quality documentation.

Kenya National Treasury

[21/01/2021 – 20/01/2022]

Data clerk

Supported records and information management activities by maintaining accurate documentation, ensuring data integrity, and improving accessibility of organisational records.

- Maintained, organised, and updated physical and digital records to ensure accuracy, accessibility, and compliance with organisational standards.
- Managed data entry, filing, retrieval, and archiving processes while safeguarding the confidentiality of sensitive information.
- Monitored data quality and conducted routine audits to verify the accuracy and consistency of records.
- Compiled, analysed, and prepared reports using operational and records data to support decision-making and reporting requirements.
- Coordinated documentation workflows across departments to improve efficiency and information accessibility.
- Assisted in the implementation and continuous improvement of records management procedures and documentation practices.
- Collaborated with cross-functional teams to streamline information management processes and enhance operational effectiveness.

VOLUNTEERING

[26/08/2025 – 07/03/2026]

Chair IEEE Robotics and Automation Society Kenya Section

Provided leadership and strategic direction for national robotics initiatives, coordinated activities across student branches, and strengthened collaboration between academia, industry, and the global IEEE RAS community

[17/05/2024 – 29/08/2025] **Chair IEEE DeKUT Robotics and Automation Society IEEE RAS DEKUT Student Chapter**

I exercised my leadership skills as the Chair of the IEEE Robotics and Automation Society at Dedan Kimathi University of Technology. During this period, I organised several robotics initiatives and coordinated industry collaborations, effectively bridging the gap between academia and industry.

[01/10/2024 – 14/11/2025] **Sectional Student Representative**

I helped bridge the gap between the IEEE RAS Section and the student branch chapters, promoting robotics activities across the section. I also represented the section in the IEEE RAS Global Committee.

[04/08/2025 – 07/03/2026] **Occupational Health and Safety Committee**

Actively contributed to the Occupational Health and Safety Committee by performing safety audits, coordinating awareness programs, and supporting the implementation of corrective and preventative safety measures.

SKILLS

Industrial Automation | Mechatronics Systems | Artificial Intelligence and Automation in robotics | Python (computer programming) | C C++ C# programming | Experience with Robot Operating System (ROS) and Linux (ROS1 and ROS2) | mechanical systems | - Variable Frequency Drives (VFD) implementation , programming and controlling from PLC | Zenon SCADA | SCADA Automation | PLC Programmer (Siemens PLC, Allen Bradley PLC,Schneider PLC) | Micro controllers/Soldering/Sensor and actuation | Electrical wiring | Electrical Panel Design | Electrical vehicles | elctrial panel wiring | electrical wiring diagrams | Electric motor control | CAD modelling | pcb design | MATLAB | Arduino | Proteus | AutoCAD | Embedded Systems | Microsoft Word | Microsoft Office | Microsoft Excel | Machine Learning | Project Managment | Quick Leaner | build team spirit | Critical thinker | Team leader / Team player | adapt to change | Time Management, Attentiveness, Problem solving, Effective communication skills

EDUCATION AND TRAINING

B.Sc Mechatronic Engineering
Dedan Kimathi University of Technology
Siemens Mechatronic Systems Certified Engineer
Dedan Kimathi University of Technology

CERTIFICATIONS

[Siemens, 19/09/2025] **Siemens Mechatronic Systems Certification Program (SMSCP) Assistant**

Mode of learning: Research-Lab based

[COPA-DATA, 17/11/2025] **zenon Certificate | zenon 11**

[WTS FOUNDATION, 17/03/2026] **E-mobility**

A foundation on electrical vehicle technology.

[IEEE Kenya Section, 14/04/2024] **Solar Panel Installation**

Mode of learning: Project based

HONOURS AND AWARDS

- [19/12/2025]

MARVERICK AWARD Awarding institution: Synkron International/Adaptis

Recognizing the employee who consistently exceeds expectations and embodies the Adaptis culture of "Above and Beyond", Always.
- [16/08/2024]

Gold Medallist - Mechatronics | WorldSkills Kenya 2024 Awarding institution: WorldSkills Kenya

Secured first place in the Kenyan National WorldSkills competition under industrial automation and mechatronics.
- [23/10/2025]

IEEE Robotics and Automation Certificate of IDEA recognition Awarding institution: IEEE ROBOTICS AND AUTOMATION SOCIETY

I got a certificate of the most outstanding and active contribution to IEEE Robotics and Automation Society globally; this was due to my technical contribution to the robotics and automation community.
- [06/07/2024]

IoT development Hackathon Winner Awarding institution: HUBiquitous H2020

Completed an intensive IoT training course culminating in a competitive hackathon, where I emerged in 1st place."

CONFERENCES AND SEMINARS

- [18/10/2025 – 24/10/2025]

2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)

China

Supported logistics and conference operations at one of the world’s premier robotics research conferences, facilitating sessions and interacting with global robotics experts.

LANGUAGE SKILLS

Other language(s):

English

LISTENING C2 READING C2 WRITING C2

SPOKEN PRODUCTION C2 SPOKEN INTERACTION C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HOBBIES AND INTERESTS

Tinkering

I enjoy taking apart electronic boards and machines, fixing broken electrical equipment and building up new machines based on my imagination.

Music

I play the violin regularly to build discipline, creativity, and focused practice habits.

Sport

I enjoy playing football and lawn tennis on the weekends.

RECOMMENDATIONS

Name: Dishon Avisah Line manager

Joshua demonstrates strong technical ability, clear communication, and excellent problem-solving skills. He consistently meets project expectations with minimal supervision.

Phone number: (+254) 717287112 | **Email:** dishonavisah@gmail.com

Name: Titus Mulembo University Professor

I have known Mr Nyandwaki for several years; he is disciplined and very skilled. I strongly recommend

Phone number: (+254) 113969223 | **Email:** titus.mulembo@dkut.ac.ke

Name: Kenneth Kibe Project Manager

Mr Nyandwaki is a reliable and skilled engineer who consistently delivers high-quality work. He learns quickly, works well under pressure, and takes full ownership of his tasks

Phone number: (+254) 705218040 | **Email:** kibethuku@gmail.com

Name: Michael Gichane University Professor

I have known Joshua throughout his time at the university; he is industrious, disciplined and highly skilled. I strongly recommend

Phone number: (+254) 713933535 | **Email:** mgmureithi@gmail.com